

## ▼ Practical 1

```
import numpy as np

# 1D NumPy array
arr = np.array([10, 20, 30, 40, 50])

# Shape of the array
print("Shape of the array:", arr.shape)

# Sum of elements
print("Sum of elements:", np.sum(arr))

# Mean of elements
print("Mean of elements:", np.mean(arr))
```

```
Shape of the array: (5,)
Sum of elements: 150
Mean of elements: 30.0
```

```
import numpy as np

# 3x3 NumPy matrix
matrix = np.array([[1, 2, 1], [3, 4, 6], [2, 8, 7]])

# Transpose of the matrix
transpose_matrix = matrix.T
print("Transpose of the matrix:\n", transpose_matrix)

# Multiply matrix by 3
multiplied_matrix = matrix * 3
print("Matrix multiplied by 3:\n", multiplied_matrix)
```

```
Transpose of the matrix:
[[1 3 2]
 [2 4 8]
 [1 6 7]]
Matrix multiplied by 3:
[[ 3  6  3]
 [ 9 12 18]
 [ 6 24 21]]
```

```
import numpy as np
import pandas as pd

data = {
    'Name' : ['Vanshika', 'Shrushti', 'Dhruv', 'Bhavika', 'Aarya'],
    'Marks' : [45, 67, 33, 79, 84],
    'City' : ['Pune', 'Jaipur', 'Nagpur', 'Mumbai', 'Kolhapur']
}

# Displaying Name column
print("\n'Name' column:", data['Name'])

# Display students scoring more than 60 marks
print("\nStudents scoring more than 60 marks:")
for i in range(len(data['Marks'])):
    if data['Marks'][i] > 60:
        print(data['Name'][i])
```

```
'Name' column: ['Vanshika', 'Shrushti', 'Dhruv', 'Bhavika', 'Aarya']

Students scoring more than 60 marks:
Shrushti
Bhavika
Aarya
```

## ▼ Practical 2

```
# List of 5 numbers
list = [45, 78, 23, 12, 89]

# Append a new number to the list
list.append(33)
print("List after appending a new number:", list)

# Insert a number at index 2 in the list
```

```
list.insert(2, 65)
print("List after inserting a number at index 2:", list)

# Remove the last element of the list
list.pop()
print("List after removing the last element:", list)

print("Final list:",list)

List after appending a new number: [45, 78, 23, 12, 89, 33]
List after inserting a number at index 2: [45, 78, 65, 23, 12, 89, 33]
List after removing the last element: [45, 78, 65, 23, 12, 89]
Final list: [45, 78, 65, 23, 12, 89]
```

```
# Dictionary containing student data (Name, Age, Branch)
student_data = {
    'Name': ['Ansh', 'Kunal', 'Shruti', 'Rudra', 'Dev'],
    'Age': [20,21,19,20,22],
    'Branch': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE']
}

# Add a new key Marks
student_data['Marks'] = [85, 90, 78, 92, 88]
print("Dictionary after adding 'Marks' key:", student_data)

# Update the value of Age
updated_age = 21
student_data['Age'] = updated_age
print("Dictionary after updating 'Age' value:", student_data)

# Print all keys and values
print("\nAll keys and values:")
for key, value in student_data.items():
    print(key, ":", value)
```

Dictionary after adding 'Marks' key: {'Name': ['Ansh', 'Kunal', 'Shruti', 'Rudra', 'Dev'], 'Age': [20, 21, 19, 20, 22], 'Branch': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE'], 'Marks': [85, 90, 78, 92, 88]}

Dictionary after updating 'Age' value: {'Name': ['Ansh', 'Kunal', 'Shruti', 'Rudra', 'Dev'], 'Age': 21, 'Branch': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE'], 'Marks': [85, 90, 78, 92, 88]}

All keys and values:  
Name : ['Ansh', 'Kunal', 'Shruti', 'Rudra', 'Dev']  
Age : 21  
Branch : ['CSE', 'ECE', 'CSE', 'ECE', 'CSE']  
Marks : [85, 90, 78, 92, 88]

```
# Create a set of numbers
numbers = {11,55,88,33,99}

# Add a new element
numbers.add(77)
print("Set after adding a new element:", numbers)

# Remove an element
numbers.remove(33)
print("Set after removing an element:", numbers)

# Print the final set
print("Final set:", numbers)

Set after adding a new element: {33, 99, 55, 88, 11, 77}
Set after removing an element: {99, 55, 88, 11, 77}
Final set: {99, 55, 88, 11, 77}
```

## ▼ Practical 3

```
import seaborn as sns

# Load the Titanic dataset using Seaborn
titanic_df = sns.load_dataset('titanic')

# Display the Titanic dataset
print(titanic_df)

# Extract the age column as a NumPy array after removing missing values
age_array = titanic_df['age'].dropna().values

# Display the age column
print("Age column:")
print(age_array)

# First 10 values
print("First 10 values:")
```

```

print(age_array[:10])

# Shape of the array
print("Shape of the array:", age_array.shape)

# Create a multi-dimensional NumPy array using columns
multi_dimensional_array = titanic_df[['age', 'fare', 'sibsp', 'parch']].values

# Display the multi-dimensional array
print("Multi-dimensional array:")
print(multi_dimensional_array)

# Shape of array
print("Shape of the array:", multi_dimensional_array.shape)

```

|     | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class  | \ |
|-----|----------|--------|--------|------|-------|-------|---------|----------|--------|---|
| 0   | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third  |   |
| 1   | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First  |   |
| 2   | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third  |   |
| 3   | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First  |   |
| 4   | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | Third  |   |
| ..  | ...      | ...    | ...    | ...  | ...   | ...   | ...     | ...      | ...    |   |
| 886 | 0        | 2      | male   | 27.0 | 0     | 0     | 13.0000 | S        | Second |   |
| 887 | 1        | 1      | female | 19.0 | 0     | 0     | 30.0000 | S        | First  |   |
| 888 | 0        | 3      | female | NaN  | 1     | 2     | 23.4500 | S        | Third  |   |
| 889 | 1        | 1      | male   | 26.0 | 0     | 0     | 30.0000 | C        | First  |   |
| 890 | 0        | 3      | male   | 32.0 | 0     | 0     | 7.7500  | Q        | Third  |   |

|     | who   | adult_male | deck | embark_town | alive | alone |
|-----|-------|------------|------|-------------|-------|-------|
| 0   | man   | True       | NaN  | Southampton | no    | False |
| 1   | woman | False      | C    | Cherbourg   | yes   | False |
| 2   | woman | False      | NaN  | Southampton | yes   | True  |
| 3   | woman | False      | C    | Southampton | yes   | False |
| 4   | man   | True       | NaN  | Southampton | no    | True  |
| ..  | ...   | ...        | ...  | ...         | ...   | ...   |
| 886 | man   | True       | NaN  | Southampton | no    | True  |
| 887 | woman | False      | B    | Southampton | yes   | True  |
| 888 | woman | False      | NaN  | Southampton | no    | False |
| 889 | man   | True       | C    | Cherbourg   | yes   | True  |
| 890 | man   | True       | NaN  | Queenstown  | no    | True  |

[891 rows x 15 columns]

Age column:

|      |      |      |      |      |      |      |      |     |     |     |      |
|------|------|------|------|------|------|------|------|-----|-----|-----|------|
| 22.  | 38.  | 26.  | 35.  | 35.  | 54.  | 2.   | 27.  | 14. | 4.  | 58. | 20.  |
| 39.  | 14.  | 55.  | 2.   | 31.  | 35.  | 34.  | 15.  | 28. | 8.  | 38. | 19.  |
| 40.  | 66.  | 28.  | 42.  | 21.  | 18.  | 14.  | 40.  | 27. | 3.  | 19. | 18.  |
| 7.   | 21.  | 49.  | 29.  | 65.  | 21.  | 28.5 | 5.   | 11. | 22. | 38. | 45.  |
| 4.   | 29.  | 19.  | 17.  | 26.  | 32.  | 16.  | 21.  | 26. | 32. | 25. | 0.83 |
| 30.  | 22.  | 29.  | 28.  | 17.  | 33.  | 16.  | 23.  | 24. | 29. | 20. | 46.  |
| 26.  | 59.  | 71.  | 23.  | 34.  | 34.  | 28.  | 21.  | 33. | 37. | 28. | 21.  |
| 38.  | 47.  | 14.5 | 22.  | 20.  | 17.  | 21.  | 70.5 | 29. | 24. | 2.  | 21.  |
| 32.5 | 32.5 | 54.  | 12.  | 24.  | 45.  | 33.  | 20.  | 47. | 29. | 25. | 23.  |
| 19.  | 37.  | 16.  | 24.  | 22.  | 24.  | 19.  | 18.  | 19. | 27. | 9.  | 36.5 |
| 42.  | 51.  | 22.  | 55.5 | 40.5 | 51.  | 16.  | 30.  | 44. | 40. | 26. | 17.  |
| 1.   | 9.   | 45.  | 28.  | 61.  | 4.   | 1.   | 21.  | 56. | 18. | 50. | 30.  |
| 36.  | 9.   | 1.   | 4.   | 45.  | 40.  | 36.  | 32.  | 19. | 19. | 3.  | 44.  |
| 58.  | 42.  | 24.  | 28.  | 34.  | 45.5 | 18.  | 2.   | 32. | 26. | 16. | 40.  |
| 24.  | 35.  | 22.  | 30.  | 31.  | 27.  | 42.  | 32.  | 30. | 16. | 27. | 51.  |
| 38.  | 22.  | 19.  | 20.5 | 18.  | 35.  | 29.  | 59.  | 5.  | 24. | 44. | 8.   |
| 19.  | 33.  | 29.  | 22.  | 30.  | 44.  | 25.  | 24.  | 37. | 54. | 29. | 62.  |
| 30.  | 41.  | 29.  | 30.  | 35.  | 50.  | 3.   | 52.  | 40. | 36. | 16. | 25.  |
| 58.  | 35.  | 25.  | 41.  | 37.  | 63.  | 45.  | 7.   | 35. | 65. | 28. | 16.  |
| 19.  | 33.  | 30.  | 22.  | 42.  | 22.  | 26.  | 19.  | 36. | 24. | 24. | 23.5 |
| 2.   | 50.  | 19.  | 0.92 | 17.  | 30.  | 30.  | 24.  | 18. | 26. | 28. | 43.  |
| 26.  | 24.  | 54.  | 31.  | 40.  | 22.  | 27.  | 30.  | 22. | 36. | 61. | 36.  |
| 31.  | 16.  | 45.5 | 38.  | 16.  | 29.  | 41.  | 45.  | 45. | 2.  | 24. | 28.  |
| 25.  | 36.  | 24.  | 40.  | 3.   | 42.  | 23.  | 15.  | 25. | 28. | 22. | 38.  |
| 40.  | 29.  | 45.  | 35.  | 30.  | 60.  | 24.  | 25.  | 18. | 19. | 22. | 3.   |
| 22.  | 27.  | 20.  | 19.  | 42.  | 1.   | 32.  | 35.  | 18. | 1.  | 36. | 17.  |
| 36.  | 21.  | 28.  | 23.  | 24.  | 22.  | 31.  | 46.  | 23. | 28. | 39. | 26.  |
| 21.  | 28.  | 20.  | 34.  | 51.  | 3.   | 21.  | 33.  | 44. | 34. | 18. | 30.  |
| 10.  | 21.  | 29.  | 28.  | 18.  | 28.  | 19.  | 32.  | 28. | 42. | 17. | 50.  |
| 14.  | 21.  | 24.  | 64.  | 31.  | 45.  | 20.  | 25.  | 28. | 4.  | 13. | 34.  |

```

# Create a new DataFrame from the multi-dimensional array
df_from_array = pd.DataFrame(multi_dimensional_array, columns=['age', 'fare', 'sibsp', 'parch'])

# Display the first 5 rows of the new DataFrame
display(df_from_array.head())

```

|   | age  | fare    | sibsp | parch |  |
|---|------|---------|-------|-------|---------------------------------------------------------------------------------|
| 0 | 22.0 | 7.2500  | 1.0   | 0.0   |                                                                                 |
| 1 | 38.0 | 71.2833 | 1.0   | 0.0   |                                                                                 |
| 2 | 26.0 | 7.9250  | 0.0   | 0.0   |                                                                                 |
| 3 | 35.0 | 53.1000 | 1.0   | 0.0   |                                                                                 |
| 4 | 35.0 | 8.0500  | 0.0   | 0.0   |                                                                                 |